

A Catalan Classification of Extremal Even-Intersecting Lattice Paths

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Abstract

Shankar proved that a family of North-East lattice paths from $(0,0)$ to (n,n) in which every two distinct paths have an even number of common edges has size at most 2^n . He constructed C_n extremal families from the noncrossing perfect matchings of $[2n]$ and conjectured that these are all the extremal families. This note proves the conjecture. The main step is an equality classification for the quadratic lemma underlying Shankar's upper bound: its equality cases are indexed by Dyck prefixes. Two auxiliary rigidity statements, one for these Dyck-prefix partitions and one for the parity kernel on the Boolean cube, then determine the extremal relation completely.

Keywords. lattice paths, extremal set theory, Eventown, Catalan numbers, noncrossing matchings.

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1 Introduction

A North-East lattice path from $(0,0)$ to (n,n) is a word in $\{E, N\}^{2n}$ with n copies of each letter. Two such paths are *even-intersecting* if the number of common lattice edges is even. Shankar [1] proved that every pairwise even-intersecting family has size at most 2^n . For a noncrossing perfect matching M of $[2n]$, he also considered

$$\mathcal{F}_M = \{w \in \{E, N\}^{2n} : \{w_i, w_j\} = \{E, N\} \text{ for every } \{i, j\} \in M\}. \quad (1)$$

Each $\mathcal{F}_M$ has size 2^n and is pairwise even-intersecting [1, Section 4]. Since the noncrossing perfect matchings of $[2n]$ are counted by the Catalan number C_n (see, for example, [2]), this gives at least C_n extremal families. Shankar conjectured that there are no others [1, Conjecture 4.1].

The purpose of this note is to prove the conjecture in its stronger structural form.

Theorem 1.1. *Let $n \geq 1$, and let $\mathcal{F}$ be a family of lattice paths from $(0,0)$ to (n,n) such that every two distinct members have an even number of common edges. If $|\mathcal{F}| = 2^n$, then there is a unique noncrossing perfect matching M of $[2n]$ for which*

$$\mathcal{F} = \mathcal{F}_M.$$

Consequently there are exactly C_n extremal families.

The proof follows the equality structure in Shankar's argument rather than introducing a different upper bound. It has three ingredients. First, equality in the quadratic lemma from [1] is classified by Dyck prefixes. Second, two such equality partitions cannot be cross-compatible unless their Dyck prefixes have the same matched edges. Third, a weight-preserving automorphism of the Boolean cube that preserves the path-intersection parity kernel is trivial. The essential point is to retain the full equality information throughout the induction.

2 The parity kernel and Dyck-prefix partitions

Identify an m -step path with a word $x = x_1 \cdots x_m \in \{0, 1\}^m$, where $1 = N$ and $0 = E$. Write $\text{wt}(x) = \sum_i x_i$. Following [1], put

$$B_m(x, y) = |E(x) \cap E(y)| \pmod{2} \in \mathbb{F}_2.$$

Equivalently,

$$B_m(x, y) = \sum_{i=1}^m \mathbf{1}_{\{x_i=y_i\}} \mathbf{1}_{\{\sum_{j<i} x_j = \sum_{j<i} y_j\}} \quad \text{in } \mathbb{F}_2. \quad (2)$$

In particular $B_m(x, x) = m \pmod{2}$.

The last coordinate gives the basic recurrence. If $p, q \in \{0, 1\}^{m-1}$ and $\alpha, \beta \in \{0, 1\}$, then

$$B_m(p\alpha, q\beta) = B_{m-1}(p, q) + \mathbf{1}_{\{\alpha=\beta\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(q)\}}. \quad (3)$$

2.1 Dyck-prefix blocks

A *Dyck prefix* of length m is a word $P = P_1 \cdots P_m$ in the alphabet $\{U, D\}$ whose height never becomes negative. Match every D to the most recent unmatched U . Let $\mathcal{E}(P)$ be the resulting noncrossing partial matching on $[m]$, and let

$$u_1 < \cdots < u_r$$

be the unmatched positions. If $k = |\mathcal{E}(P)|$, then $m = 2k + r$.

For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r) \in \{0, 1\}^r$ define

$$\mathcal{A}_\varepsilon(P) = \{x \in \{0, 1\}^m : x_a \neq x_b \text{ for every } \{a, b\} \in \mathcal{E}(P), \quad x_{u_i} = \varepsilon_i \ (1 \leq i \leq r)\}. \quad (4)$$

Thus every block has size 2^k and lies in the weight level $k + \text{wt}(\varepsilon)$. Put

$$\Omega(P) = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P).$$

The next cancellation fact is the reason these blocks appear naturally.

Lemma 2.1 (cancellation). *Let P be a Dyck prefix of length m . Suppose $x, y \in \Omega(P)$, and delete from both words every coordinate belonging to an edge of $\mathcal{E}(P)$. If the resulting words are $\bar{x}, \bar{y} \in \{0, 1\}^r$, then*

$$B_m(x, y) = B_r(\bar{x}, \bar{y}). \quad (5)$$

In particular, if $x \in \mathcal{A}_\varepsilon(P)$ and $y \in \mathcal{A}_\eta(P)$, then

$$B_m(x, y) = B_r(\varepsilon, \eta). \quad (6)$$

Proof. No unmatched position of a Dyck prefix can lie strictly inside a matched edge: otherwise it would still be on top of the stack when that edge closed. Hence every nonempty matching $\mathcal{E}(P)$ contains an innermost edge $\{i, i+1\}$. Both x and y use one 0 and one 1 at these two positions. If the orientations agree, the two positions contribute either two common edges or none; if the orientations disagree, they contribute none. Hence their contribution is even. After the two positions are deleted, both paths have been translated by the same vector $(1, 1)$ from that point onward, so all later common-edge incidences are unchanged. Deleting innermost edges successively proves (5). $\square$

It follows at once that the blocks (4) satisfy the three compatibility conditions that will occur in Section 3: within a block the value of B_m is $r \equiv m \pmod{2}$, while on a product of two blocks it is constant.

We shall also need a two-step test. For $s \geq 1$, write a word of length $s + 1$ as $z\alpha\beta$ with $z \in \{0, 1\}^{s-1}$.

Lemma 2.2 (square test). *For $y = y'\beta\gamma \in \{0, 1\}^{s+1}$, the following are equivalent:*

- (i) $B_{s+1}(z10, y) = B_{s+1}(z01, y)$ for every $z \in \{0, 1\}^{s-1}$;
- (ii) $\beta \neq \gamma$.

Proof. Let $h = \text{wt}(z)$ and $q = \text{wt}(y')$. Applying (3) twice gives

$$\begin{aligned} B_{s+1}(z10, y'\beta\gamma) + B_{s+1}(z01, y'\beta\gamma) \\ = \mathbf{1}_{\{h=q\}} + \mathbf{1}_{\{\gamma=0\}}\mathbf{1}_{\{h=q+\beta-1\}} + \mathbf{1}_{\{\gamma=1\}}\mathbf{1}_{\{h=q+\beta\}}. \end{aligned}$$

If $\beta \neq \gamma$, the right-hand side is identically zero. If $\beta = \gamma = 0$, take $h = q$; if $\beta = \gamma = 1$, again take $h = q$. Since $0 \leq q \leq s - 1$, such a word z exists, and the displayed value is then 1. $\square$

2.2 Cross-rigidity

For Dyck prefixes P, Q of the same length, write $P \preccurlyeq Q$ if, for every block A of the partition $\{\mathcal{A}_\varepsilon(P)\}$, every $x, x' \in A$, and every $y \in \Omega(Q)$,

$$B_m(x, y) = B_m(x', y). \tag{7}$$

Thus changing the orientations of the matched edges of P is invisible to every word satisfying the matching constraints of Q .

Lemma 2.3. *For Dyck prefixes P, Q of length m ,*

$$P \preccurlyeq Q \implies \mathcal{E}(P) \subseteq \mathcal{E}(Q).$$

Proof. Proceed by induction on m . Remove the last symbol from P and Q , obtaining Dyck prefixes P^- and Q^- . Every word in $\Omega(Q^-)$ can be extended to a word in $\Omega(Q)$: if the last symbol of Q is U , choose either last bit; if it is D , the last bit is forced to be opposite to the bit at its matched opener. Likewise, any two words in the same block for P^- can be extended with a common last bit to two words in one block for P ; when the last symbol of P is D , the bit at its opener is fixed inside a block of P^- . Since all words in one block have the same weight, the correction term in (3) is the same for the two extensions. Hence

$$P^- \preccurlyeq Q^-.$$

By induction, $\mathcal{E}(P^-) \subseteq \mathcal{E}(Q^-)$.

If the last symbol of P is U , this already gives $\mathcal{E}(P) \subseteq \mathcal{E}(Q)$. Suppose instead that it is D , and let e be its matched edge. Delete the endpoints of all edges in $\mathcal{E}(P^-)$. By induction these edges also belong to $\mathcal{E}(Q^-)$. Lemma 2.1, applied successively to the same matched pairs on both sides of (7), shows that every defining B -equality is unchanged; hence $\preccurlyeq$ descends to the reduced prefixes. Removing the endpoints of a matched U – D pair preserves the Dyck-prefix property. On the P -side this is equivalently the cancellation associated with P^-U , whose matching is exactly $\mathcal{E}(P^-)$. The reduced prefix coming from P has the form

$$U^s D$$

for some $s \geq 1$: all surviving vertices before the final D are unmatched, and the last of them is its opener. Let Q' be the reduced prefix coming from Q .

A block of $U^s D$ is $\{z10, z01\}$ for some $z \in \{0, 1\}^{s-1}$. Thus, for every $y \in \Omega(Q')$,

$$B_{s+1}(z10, y) = B_{s+1}(z01, y) \quad \text{for all } z \in \{0, 1\}^{s-1}.$$

Lemma 2.2 gives $y_s \neq y_{s+1}$ for every $y \in \Omega(Q')$. In a Dyck-prefix matching, two positions have opposite bits in every word of $\Omega(Q')$ if and only if they are matched to one another: if they lie on different matching edges, or if one of them is unmatched, the relevant orientations and unmatched bits can be chosen independently to make the two bits equal. Therefore $\{s, s+1\} \in \mathcal{E}(Q')$. Lifting this edge back gives $e \in \mathcal{E}(Q)$. $\square$

Proposition 2.4 (cross-rigidity). *Let P, Q be Dyck prefixes of length m . Suppose that B_m is constant on*

$$\mathcal{A}_\varepsilon(P) \times \mathcal{A}_\eta(Q)$$

for every pair of labels ε, η . Then $P = Q$.

Proof. The hypothesis gives $P \preceq Q$. Since B_m is symmetric, it also gives $Q \preceq P$. Lemma 2.3 yields $\mathcal{E}(P) = \mathcal{E}(Q)$. A Dyck prefix is determined by its matching edges: the right endpoints are the D -positions, and every other position is a U -position. Hence $P = Q$. $\square$

3 Equality in the quadratic lemma

Shankar's proof of the even-intersection bound rests on the following inequality [1, Lemma 2.3]. Empty members do not affect the inequality and may be discarded. The equality classification is stated simultaneously.

Theorem 3.1 (quadratic equality classification). *Let $A_1, \dots, A_t$ be nonempty pairwise disjoint subsets of $\{0, 1\}^m$ such that*

- (i) *each A_i is contained in one Hamming-weight level;*
- (ii) *$B_m(x, y) = m \pmod{2}$ for all $x, y \in A_i$;*
- (iii) *for $i \neq j$, the function B_m is constant on $A_i \times A_j$.*

Then

$$\sum_{i=1}^t |A_i|^2 \leq 2^m. \tag{8}$$

Equality holds if and only if there is a Dyck prefix P of length m such that, up to reordering,

$$\{A_1, \dots, A_t\} = \{\mathcal{A}_\varepsilon(P) : \varepsilon \in \{0, 1\}^r\},$$

where r is the number of unmatched positions of P . The prefix P is unique.

The equality analysis reduces to the following local Boolean-cube classification. The same lemma will later control automorphisms.

For $x, y \in \{0, 1\}^d$, set

$$\Delta_{x,y}(z) = B_d(x, z) + B_d(y, z) \in \mathbb{F}_2.$$

Let $\mathcal{L}_j^{(d)} = \{z \in \{0, 1\}^d : \text{wt}(z) = j\}$, with the convention that this set is empty if $j \notin \{0, \dots, d\}$.

Lemma 3.2 (localized pairs). *Let $u, v \in \{0, 1\}^d$ satisfy $\text{wt}(u) = k + 1$ and $\text{wt}(v) = k$. Assume*

$$\Delta_{u,v}(z) = 0 \quad \text{whenever } \text{wt}(z) \notin \{k, k + 1\}. \quad (9)$$

Then either

$$u = t1, \quad v = t0 \quad (10)$$

for some $t \in \{0, 1\}^{d-1}$, or $d = 2k + 1 \geq 3$ and

$$u = 1(10)^k, \quad v = 0(01)^k. \quad (11)$$

For the exceptional pair,

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z) \in \{k, k + 1\}. \quad (12)$$

Proof. Write $u = p\alpha$, $v = q\beta$, and $z\gamma$ for the test word. From (3),

$$\Delta_{p\alpha, q\beta}(z\gamma) = \Delta_{p,q}(z) + \mathbf{1}_{\{\alpha=\gamma\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(z)\}} + \mathbf{1}_{\{\beta=\gamma\}} \mathbf{1}_{\{\text{wt}(q)=\text{wt}(z)\}}. \quad (13)$$

The proof is a finite-state induction; the details below account for the exceptional alternating state.

First consider the following auxiliary states.

- $S_d(k)$: $\text{wt}(x) = \text{wt}(y) = k$ and $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$;
- $E_d(k)$: $\text{wt}(x) = \text{wt}(y) = k$ and $\Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)}}$;
- $L_d^-(k)$: $\text{wt}(x) = k + 1$, $\text{wt}(y) = k$, and $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$;
- $L_d^+(k)$: the same weight condition, with support contained in $\mathcal{L}_{k+1}^{(d)}$.

The case $d = 0$ is immediate. A simultaneous induction on d , using (13), gives

$$S_d(k) \implies x = y, \quad E_d(k), L_d^-(k), L_d^+(k) \text{ are impossible.} \quad (14)$$

The transition rules are as follows. A terminal pattern whose implied prefix weight lies outside $\{0, \dots, d - 1\}$ is infeasible; whenever a pattern is feasible, each weight level used in the argument below is nonempty. In the state $E_d(k)$, equal terminal bits are impossible because the left side of (13) is then independent of γ while the target value changes between two consecutive weight layers; unequal terminal bits force $\Delta_{p,q} \equiv 0$ for a pair of adjacent weights, contradicting the induction hypothesis for L^-/L^+ . For $L_d^-(k)$, the terminal patterns 00 and 01 give an immediate contradiction by evaluating (13) on a word of weight $k + 1$; the pattern 10 reduces to $E_{d-1}(k)$, while 11 reduces to $L_{d-1}^+(k - 1)$. If $\bar{x}$ denotes the bitwise complement of x , then $B_d(\bar{x}, \bar{z}) = B_d(x, z)$. Hence simultaneous complementation of the two words and the test word, followed by swapping the first two words, sends an L^+ state to an L^- state. Finally, in $S_d(k)$, equal terminal bits force $\Delta_{p,q} \equiv 0$, so induction gives $p = q$; unequal terminal bits again force an adjacent-weight pair with zero profile, contradicting L^-/L^+ .

It remains to track two alternating states. Let

$$\begin{aligned} M_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k - 1, \quad \text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}, \\ T_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k, \quad \Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)} \cup \mathcal{L}_{k+1}^{(d)}}. \end{aligned}$$

The remaining two states have the following transitions, again by (13). In an $M_d(k)$ state, the patterns 00 and 01 are contradicted by testing a word of weight $k + 1$, while 11 is contradicted by testing a word of weight $k - 2$ (the pattern is feasible only when that layer is nonempty).

The only surviving pattern is 10, and the prefix then has value 1 on exactly the two layers $k-1, k$; hence it is a $T_{d-1}(k-1)$ pair. In a $T_d(k)$ state with $d > 1$, the patterns 00 and 11 are contradicted by testing a prefix word of weight k , and 10 gives the same value for the two last-bit choices although the target values differ on one of the adjacent nonempty layers. Thus only 01 survives, and (13) says that its prefix has support exactly on layer k , so it is an $M_{d-1}(k)$ pair. Since $T_1(0)$ consists only of $(1, 0)$, induction gives

$$M_{2k}(k) : (x, y) = (11(01)^{k-1}, 00(10)^{k-1}) \quad (k \geq 1), \quad (15)$$

$$T_{2k+1}(k) : (x, y) = (1(10)^k, 0(01)^k). \quad (16)$$

No M or T state occurs in the opposite parity.

Now apply (13) to a pair satisfying (9). According to the terminal pattern (α, β) , its prefix lies in the following state:

(α, β)	prefix state	consequence
00	L^-	impossible
11	L^+	impossible
10	S	$p = q$
01	M	the alternating pair (11).

The 10 case is exactly (10). The 01 case, together with (15), yields (11); its profile is (12) by (16). When $d = 1$, this alternating pair is the canonical pair itself, so the exceptional alternative is new only for $d \geq 3$. $\square$

We next isolate the reconstruction step that occurs after the two inductive halves have become the same Dyck-prefix partition.

For a family $\mathcal{C} = \{C_i\}$ of nonempty subsets of $\{0, 1\}^{r+1}$, write

$$C_i^a = \{z \in \{0, 1\}^r : za \in C_i\}, \quad a \in \{0, 1\},$$

let $I(\mathcal{C})$ be the indices for which both C_i^0 and C_i^1 are nonempty, and define

$$\begin{aligned} \mathcal{G}_0(\mathcal{C}) &= \{C_i^0 : C_i^0 \neq \emptyset\} \cup \{C_i^1 : i \in I(\mathcal{C})\}, \\ \mathcal{G}_1(\mathcal{C}) &= \{C_i^1 : C_i^1 \neq \emptyset\} \cup \{C_i^0 : i \in I(\mathcal{C})\}. \end{aligned}$$

These are the two split families used in Shankar's induction.

Lemma 3.3 (reconstruction). *Let $r \geq 0$. Suppose a family $\mathcal{C}$ of nonempty subsets of $\{0, 1\}^{r+1}$ satisfies the three conditions of Theorem 3.1, and suppose*

$$\mathcal{G}_0(\mathcal{C}) = \mathcal{G}_1(\mathcal{C}) = \{\{z\} : z \in \{0, 1\}^r\}.$$

Then exactly one of the following holds:

$$(a) \ \mathcal{C} = \{\{z0\}, \{z1\} : z \in \{0, 1\}^r\};$$

$$(b) \ r \geq 1 \text{ and}$$

$$\mathcal{C} = \{\{t10, t01\} : t \in \{0, 1\}^{r-1}\}.$$

Proof. Every nonempty half C_i^a is a singleton, since it occurs in one of the two displayed singleton partitions. There is also no overlap between opposite halves when one of the corresponding blocks is split. Indeed, suppose $i \in I(\mathcal{C})$, $j \neq i$, and

$$z \in C_j^0 \cap C_i^1.$$

Choose $x \in C_i^0$. Then $z1, x0 \in C_i$, so condition (ii) gives $B_{r+1}(z1, x0) = r + 1 \pmod{2}$. Weight homogeneity gives $\text{wt}(x) = \text{wt}(z) + 1$, and (3) therefore yields

$$B_{r+1}(z0, x0) = B_{r+1}(z1, x0) = r + 1 \pmod{2}.$$

On the other hand, $B_{r+1}(z0, z1) = B_r(z, z) = r \pmod{2}$, contradicting constancy on $C_j \times C_i$. The case $C_j^1 \cap C_i^0 \neq \emptyset$ is symmetric.

It follows that each label $z \in \{0, 1\}^r$ either contributes the two singleton blocks $\{z0\}, \{z1\}$, or occurs in exactly one split block. Such a block has the form

$$\{u0, v1\}, \quad \text{wt}(u) = \text{wt}(v) + 1, \quad (17)$$

where the weight relation follows from condition (i). The labels used in distinct two-element blocks are disjoint.

Fix a block (17), and write $k = \text{wt}(v)$. If a label z is unpaired, cross-constancy with the two singleton blocks $\{z0\}$ and $\{z1\}$ gives

$$\Delta_{u,v}(z) = \mathbf{1}_{\{\text{wt}(z)=k+1\}} = \mathbf{1}_{\{\text{wt}(z)=k\}}. \quad (18)$$

If another two-element block is $\{a0, b1\}$, comparison of the two rows in its two columns gives

$$\Delta_{u,v}(a) = \mathbf{1}_{\{\text{wt}(a)=k+1\}}, \quad \Delta_{u,v}(b) = \mathbf{1}_{\{\text{wt}(b)=k\}}. \quad (19)$$

Equations (18)–(19) imply that $\Delta_{u,v}$ vanishes outside the two levels $k, k + 1$. Lemma 3.2 therefore says that every paired block is canonical,

$$\{t10, t01\},$$

except possibly for the unique alternating pair in odd dimension.

First exclude that exception. Suppose $r = 2s + 1 \geq 3$ and

$$u = 1(10)^s, \quad v = 0(01)^s.$$

By (12) and (18)–(19), every word of weight $s + 1$ must occur as a high endpoint and every word of weight s as a low endpoint. All other pairs are canonical. A canonical high endpoint ends in 1, while a canonical low endpoint ends in 0. If $s \geq 2$, there are $\binom{2s}{s-1} > 1$ weight- s words ending in 1, so one of them is different from v ; it can be neither the exceptional low endpoint nor a canonical low endpoint, a contradiction. If $s = 1$, the exceptional block is $\{1100, 0011\}$ and one forced canonical block is $\{0110, 0101\}$. Their B_4 -matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

contrary to cross-constancy. Thus no exceptional pair occurs.

It remains to show that the canonical pairs are either absent or all present. Suppose the pair indexed by $t \in \{0, 1\}^{r-1}$ is present, and put $h = \text{wt}(t)$. For $u = t1$ and $v = t0$, recurrence (3) gives

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z_1 \cdots z_{r-1}) = h. \quad (20)$$

No label of total weight h or $h + 1$ can be unpaired by (18). Combining (20) with (19) shows that the labels

$$s0 \text{ (wt}(s) = h), \quad s1 \text{ (wt}(s) = h - 1), \quad s1 \text{ (wt}(s) = h), \quad s0 \text{ (wt}(s) = h + 1)$$

must respectively be low, high, high, and low endpoints. Hence every canonical pair whose prefix weight lies in $\{h - 1, h, h + 1\} \cap \{0, \dots, r - 1\}$ is present. Iterating along the connected sequence of weight levels $0, 1, \dots, r - 1$ forces all canonical pairs. This proves the dichotomy. $\square$

Proof of Theorem 3.1. Proceed by induction on m , proving the inequality and its equality statement simultaneously. The case $m = 0$ is immediate. For every i , write

$$A_i^a = \{x \in \{0, 1\}^{m-1} : xa \in A_i\}, \quad a \in \{0, 1\},$$

and let I be the set of indices for which both A_i^0 and A_i^1 are nonempty. Consider the two collections of nonempty sets

$$\begin{aligned} \mathcal{G}_0 &= \{A_i^0 : A_i^0 \neq \emptyset\} \cup \{A_i^1 : i \in I\}, \\ \mathcal{G}_1 &= \{A_i^1 : A_i^1 \neq \emptyset\} \cup \{A_i^0 : i \in I\}. \end{aligned}$$

Each collection satisfies the three hypotheses in dimension $m - 1$. The weight condition is immediate, and (3) transfers internal and cross-constancy. It remains to verify disjointness. Suppose, for example, that

$$u \in A_j^0 \cap A_i^1$$

with $i \in I$. If $i = j$, then $u0, u1 \in A_i$ but $B_m(u0, u1) = m - 1$, contradicting (ii). Thus $i \neq j$. Choose $x \in A_i^0$. Then $u1, x0 \in A_i$, so $B_m(u1, x0) = m$. Since these two words have the same total weight, $\text{wt}(x) = \text{wt}(u) + 1$, and (3) gives

$$B_m(u0, x0) = B_m(u1, x0) = m.$$

On the other hand $B_m(u0, u1) = m - 1$. This contradicts constancy on $A_j \times A_i$. The other possible overlap is symmetric.

Put $a_i = |A_i^0|$ and $b_i = |A_i^1|$. By induction,

$$\sum_{i \notin I} a_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}, \quad (21)$$

$$\sum_{i \notin I} b_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}. \quad (22)$$

For $i \in I$,

$$(a_i + b_i)^2 \leq 2a_i^2 + 2b_i^2,$$

with equality exactly when $a_i = b_i$. Adding (21) and (22) proves (8).

Assume now that equality holds. Then both (21) and (22) are equalities and $a_i = b_i$ for every $i \in I$. By induction there are Dyck prefixes P, Q of length $m - 1$ such that $\mathcal{G}_0$ and $\mathcal{G}_1$ are their block partitions. Moreover B_{m-1} is constant on every block of $\mathcal{G}_0$ times every block of $\mathcal{G}_1$: if the two prefix blocks come from A_i^α and A_j^β , this follows from (3), since its correction term is constant on that product and B_m is constant on $A_i \times A_j$ (also when $i = j$). Proposition 2.4 yields $P = Q$.

Delete from every prefix word the coordinates belonging to $\mathcal{E}(P)$. Equivalently, apply Lemma 2.1 to the Dyck prefix PU , whose last position is unmatched; this preserves all B -conditions, including the final bit of the original words. Since every block of P collapses to one label in $\{0, 1\}^r$, the reduced reconstruction is exactly the situation of Lemma 3.3. In case (a) the new Dyck prefix is PU ; in case (b) it is PD . This completes the induction and proves existence. Conversely, if P has k matched edges and r unmatched positions, Lemma 2.1 shows that its blocks satisfy (i)–(iii), and

$$\sum_{\varepsilon \in \{0, 1\}^r} |\mathcal{A}_\varepsilon(P)|^2 = 2^r (2^k)^2 = 2^{2k+r} = 2^m.$$

Thus every Dyck-prefix partition is indeed an equality case.

For uniqueness, note that from the union $\Omega(P)$ one recovers the matching by

$$\{i, j\} \in \mathcal{E}(P) \iff x_i \neq x_j \text{ for every } x \in \Omega(P).$$

If i, j are not matched, their bits can be made equal by independently orienting their matching edges and choosing unmatched bits. Hence the equality partition determines P . $\square$

4 Rigidity on the Boolean cube

The final equality step produces a bijection of the unmatched labels. The next proposition shows that this bijection cannot be nontrivial.

Proposition 4.1 (cube rigidity). *Let $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$ be a bijection satisfying*

$$\text{wt}(\phi(x)) = \text{wt}(x), \quad B_r(\phi(x), \phi(y)) = B_r(x, y) \quad (23)$$

for all $x, y \in \{0, 1\}^r$. Then ϕ is the identity.

Proof. For words of adjacent weights, define an intrinsic relation

$$u \sim v \iff \text{wt}(u) = \text{wt}(v) + 1 \text{ and } \Delta_{u,v}(z) = 0 \text{ outside the two levels } \text{wt}(v), \text{wt}(u).$$

Lemma 3.2 identifies all edges of this relation. They are the canonical edges

$$t1 \sim t0 \quad (t \in \{0, 1\}^{r-1}), \quad (24)$$

plus, when $r \geq 3$ is odd, the single exceptional edge

$$1(10)^{(r-1)/2} \sim 0(01)^{(r-1)/2}.$$

The relation is preserved by ϕ because both weight and B_r are preserved.

If r is even, (24) is therefore preserved as a set. If $r \geq 3$ is odd, the two endpoints of the exceptional edge are the only vertices of degree two in the relation graph; hence the exceptional edge is intrinsically distinguished, and after removing it the canonical matching is again preserved. The cases $r = 0, 1$ are immediate. Thus in all cases ϕ maps every canonical pair to a canonical pair. Weight preservation prevents the two endpoints from being exchanged, so there is a weight-preserving bijection $\psi : \{0, 1\}^{r-1} \rightarrow \{0, 1\}^{r-1}$ such that

$$\phi(t\alpha) = \psi(t)\alpha \quad (t \in \{0, 1\}^{r-1}, \alpha \in \{0, 1\}).$$

Since the last bits differ,

$$B_r(t0, s1) = B_{r-1}(t, s).$$

Using (23),

$$B_{r-1}(\psi(t), \psi(s)) = B_r(\phi(t0), \phi(s1)) = B_r(t0, s1) = B_{r-1}(t, s).$$

Hence ψ satisfies the same hypotheses in dimension $r - 1$. Induction gives $\psi = \text{id}$ and therefore $\phi = \text{id}$. $\square$

5 Extremal relations

Call a relation $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$ *admissible* if

$$\text{wt}(x) = \text{wt}(y) \quad ((x, y) \in R) \quad (25)$$

and

$$B_n(x, x') = B_n(y, y') \quad \text{for all } (x, y), (x', y') \in R. \quad (26)$$

Shankar proved that every admissible relation has at most 2^n elements [1, Lemma 2.2]. Theorem 3.1 gives its complete equality case.

Proposition 5.1. *If $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$ is admissible and $|R| = 2^n$, then there are Dyck prefixes P, Q of length n , with the same number r of unmatched positions, such that*

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q). \quad (27)$$

Proof. Form the bipartite graph whose edges are the pairs in R , and ignore isolated vertices. Let A_j and B_j be the left and right vertex sets of its connected components. If two left vertices x, x' have a common neighbor y , then (26) gives

$$B_n(x, x') = B_n(y, y) = n \pmod{2}.$$

An induction on graph distance along an alternating path shows the same for any two vertices in one A_j , and similarly in one B_j : if $x_0 - y_1 - x_1 - \dots - y_\ell - x_\ell$ is a shortest alternating path, then (26) gives $B_n(x_0, x_\ell) = B_n(y_1, y_\ell)$, and the latter pair has smaller distance. Also, (25) makes each A_j and B_j weight-homogeneous. Finally, if A_i and A_j are different components, then B_n is constant on $A_i \times A_j$. Indeed, if $x, x' \in A_i$ share a neighbor y and (a, b) is any edge of the other component, then

$$B_n(x, a) = B_n(y, b) = B_n(x', a).$$

Connectivity lets one move either endpoint, so the value is constant. The same holds on the right. Thus both collections satisfy Theorem 3.1.

The standard component estimate is

$$|R| \leq \sum_j |A_j| |B_j| \leq \left(\sum_j |A_j|^2 \right)^{1/2} \left(\sum_j |B_j|^2 \right)^{1/2} \leq 2^n. \quad (28)$$

Since $|R| = 2^n$, equality holds throughout. In particular every component is complete bipartite, and both quadratic sums equal 2^n . Theorem 3.1 gives Dyck prefixes P, Q whose block partitions are $\{A_j\}$ and $\{B_j\}$. If their numbers of unmatched positions are r_P, r_Q , then the common number of components is both 2^{r_P} and 2^{r_Q} , so $r_P = r_Q =: r$. Consequently P and Q also have the same number $k = (n - r)/2$ of matched edges.

Index the blocks of P by $\varepsilon \in \{0, 1\}^r$. The components give a bijection $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$ such that

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_{\phi(\varepsilon)}(Q).$$

A word in $\mathcal{A}_\varepsilon(P)$ has weight $k + \text{wt}(\varepsilon)$, and a word in $\mathcal{A}_{\phi(\varepsilon)}(Q)$ has weight $k + \text{wt}(\phi(\varepsilon))$. Completeness of the component and (25) therefore give

$$\text{wt}(\phi(\varepsilon)) = \text{wt}(\varepsilon).$$

For labels ε, η , choose arbitrary words in the two corresponding complete components. Equation (26), followed by Lemma 2.1, gives

$$B_r(\phi(\varepsilon), \phi(\eta)) = B_r(\varepsilon, \eta).$$

Proposition 4.1 gives $\phi = \text{id}$, proving (27). □

6 Proof of the Catalan classification

Let $F = f_1 \cdots f_{2n}$ be a path from $(0, 0)$ to (n, n) , viewed as a balanced binary word, and define

$$u(F) = (f_1, \dots, f_n), \quad v(F) = (1 - f_{2n}, \dots, 1 - f_{n+1}). \quad (29)$$

Then $\text{wt}(u(F)) = \text{wt}(v(F))$. Conversely every pair (u, v) of equal weight reconstructs a unique balanced word by taking first half u and second half the reversed complement of v . Thus (29) is a bijection between paths from $(0, 0)$ to (n, n) and equal-weight pairs in $\{0, 1\}^n \times \{0, 1\}^n$.

Reversing and complementing the second half is a reflection of that half-path and preserves common edges. Therefore, for any two balanced paths F, G ,

$$B_{2n}(F, G) = B_n(u(F), u(G)) + B_n(v(F), v(G)) \quad \text{in } \mathbb{F}_2, \quad (30)$$

which is Observation 2.1 of [1].

Proof of Theorem 1.1. Let $\mathcal{F}$ be extremal and let

$$R = \{(u(F), v(F)) : F \in \mathcal{F}\}.$$

By (30), the even-intersection hypothesis says exactly that R is admissible for distinct pairs; for a pair with itself, both sides of (26) equal $n \pmod{2}$, so admissibility holds in full. Since $|R| = 2^n$, Proposition 5.1 gives Dyck prefixes P, Q of length n with r unmatched positions and

$$R = \bigcup_{\varepsilon \in \{0,1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q).$$

Write the unmatched positions as

$$a_1 < \cdots < a_r \quad \text{in } P, \quad b_1 < \cdots < b_r \quad \text{in } Q.$$

Construct a matching M on $[2n]$ as follows:

- include every edge of $\mathcal{E}(P)$ in the left half;
- reflect every edge $\{i, j\} \in \mathcal{E}(Q)$ to $\{2n + 1 - j, 2n + 1 - i\}$ in the right half;
- for $1 \leq i \leq r$, add the cross-edge

$$\{a_i, 2n + 1 - b_i\}. \quad (31)$$

This is a perfect matching. It is noncrossing: internal edges are noncrossing; the right endpoints in (31) decrease as the left endpoints increase, so the cross-edges are nested; and an unmatched position of a Dyck prefix cannot lie strictly inside one of its matched edges, since it would still be on top of the stack when that edge closed.

For an internal edge of P , membership in a block $\mathcal{A}_\varepsilon(P)$ means that the corresponding two first-half bits are opposite. The same is true for the reflected edges of Q , because complementing both endpoint bits preserves opposition. Finally, the common label ε gives

$$f_{a_i} = v(F)_{b_i} = 1 - f_{2n+1-b_i},$$

so the two bits on each cross-edge (31) are opposite. Hence every $F \in \mathcal{F}$ belongs to $\mathcal{F}_M$. Both families have size 2^n , so

$$\mathcal{F} = \mathcal{F}_M.$$

The matching is unique. Indeed, for every noncrossing perfect matching M ,

$$\{i, j\} \in M \iff w_i \neq w_j \text{ for every } w \in \mathcal{F}_M. \quad (32)$$

The forward implication is the definition. If i and j lie on different matching edges, orient those two edges independently so that $w_i = w_j$. Thus (32) recovers M from the family. Since the noncrossing perfect matchings of $[2n]$ are counted by C_n , the number of extremal families is exactly C_n . $\square$

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